Exploratory Data Analysis

Dataset: Speeddating.arff

GitHub Repo:
https://github.com/etorrey-vc/Ken-and-Friends-CMPU-250-

Google Colab for Data Analysis:
https://colab.research.google.com/drive/1ie9CqEGVOY4VrLju256nkR2bprUJr-bw

Our Research Questions

1. How do the students' preferences correlate/compare to their decisions?
2. How are different factors correlated with match success and which holds the most weight? (ambition, funniness(!), sincerity, intelligence, interests correlated)
3. Are there certain demographics (race, gender, age) that are disproportionately rated lower?
4. Are there demographics that perform well in the ratings but seem to earn fewer matches?

*** For question one maybe we can create a model that predicts a students' decision based on their evaluation of their partner but the different parameters are weighted differently based on their preferences. ***

Initial questions for us to answer

- What are our outcome variable(s), how well do they measure the outcome we are interested in, and how do they relate to our expectations? (y in the regression)
    - RQ 1: How do their preferences correlate/compare to their decisions?
        - Outcome variable(s): Student decision
            - It is a direct, behavioral measure of interest, which makes it a strong outcome for comparing what people say they want with what they actually choose. However, its limits are that it's binary, so it is not able to capture how strongly someone felt. The decision lets us test whether stated preferences hold up in practice.
            - Explanatory variable(s): Preference columns and partner evaluation columns
    - RQ 2: How are different factors correlated with decision success and which holds the most weight? (ambition, funniness(!), sincerity, intelligence, interests correlated)
        - Outcome variable: Student decision

            - Again, it is a behavioral measure of whether a date went well enough for the participant to want another that is made right after the date. It is able to directly measure our research question. In terms of our expectations, the decision lets us test which traits actually translate into a positive response from a partner.
        - Explanatory variable(s): Partner evaluation columns
    - RQ 3: Are there certain demographics (race, gender, age,) that are disproportionately rated lower?
        - Outcome Variables: Partner evaluation columns, like, match decision
            - The use of these 3 outcome variables helps measure our research question most accurately. These ratings measure perceptions, not actual traits, which is exactly what a question about biased evaluation calls for. In addition, having 3 metrics of rating can help find consistencies with disproportionate ratings. For our expectations, evaluation of traits, like, and decision let us test whether participants evaluate partners more favorably when those partners share their demographic background.
        - Explanatory variable(s): partner race, age, gender
    - RQ 4: Are there demographics that perform well in the ratings but seem to be less succesful?
        - Outcome variable(s): Student decision
            - The decision on whether the student decides to go on another date helps us test if the partner was successful or not. For our expectations, if decisions are driven entirely by how partners are rated, then race should have no extra effect once ratings are accounted for.
        - Explanatory variable(s): Evaluation of partner traits, self-evaluated trait preferences, similarity of interests, job field

Table of summary statistics

https://docs.google.com/spreadsheets/d/1Kzy503pXe1By_4ZBVWJSxsQDCMpi5WkaiNUxXGC-nh4/edit?gid=0#gid=0

Data Visualization

https://docs.google.com/document/d/1MUe-ICkGbwzM-hJ1DqZ0XtKtWpWPsUUbLRBKP3cLjfw/edit?tab=t.0

Data Preparation and Cleaning

Answer the following question:

- What data cleaning did you have to do?

1. Dropped all columns that we did not deem were necessary for our project
    a. All of the discretized columns which was a column that used values from other columns to put them in bins from ranges to sort them. We are interested in the actual values of these columns so we deemed these unnecessary
    b. All columns relating to partners evaluation of the student and partners preferences. We decided to only look at the data from the students perspective (their preferences/evaluations) because the data goes through all of the dates in both perspectives so that "partner" will be shown as the student in one data point so it was unnecessary to look at both perspectives at a single data point.
    c. All hobby columns. We decided to use the interests correlate columns that was derived from the hobby columns so we decided that it was unnecessary to keep the raw hobby columns.
    d. All columns that related to the students expectation on the dating experiment.
    e. All other miscellaneous columns that were not relevant to our project
2. Dropped all rows that had any null observations. We did this for 2 reasons:
    a. 1. Adding any sort of value to these observations would heavily skew the data especially when we train our model to predict the students decisions so we decided to omit them as it would have the least effect on our dataset.
    b. 2. It was necessary to convert all of the preference and evaluation column values to floats so that we are able to do calculations with them. If there were null values this would make it very difficult to do so.
3. Converted preference and evaluation columns to floats so we can use them for calculations including mean, median, max, minimum, etc.

- How did you prepare the data?

**For visualizations that required preparation

- Visualization #2: I averaged out decision percentages by race before making the bar chart.
- Visualization #3 : I split the data by the participant's decision, averaged each group's importance and evaluation columns, and arranged the results into two small tables (importance and rating) with one row per trait and a column for each decision group.
    - Visualization #3 (a): I divided the importance scores (points out of 100) by 10 so they're on roughly the same scale as the 1–10 ratings. Then, I averaged each column and paired it with each trait's importance with its evaluation, and sorted by importance.
    - Visualization #3 (b): Same as (a)

    - Visualization #4:
        - Visualization #4 (a): I grouped by partner race (race_o), averaged the six evaluation columns, sorted the evaluation categories by their overall average, and transposed the table so categories appear on the axis with one bar per race.
        - Visualization #4 (b): I added together all the averaged summed values for each evaluation columns
- Are you deciding to exclude any observations? If so, why?
    - Included in data cleaning processes
- Did you have to create any new variables from existing variables? If so, how and why?
    - No, for the time being